Rank filters in the quadratic memory core
Positive product channels and double-scaled tail control

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Local research note — 6 November 2026

Abstract

The previously identified finite dynamical residual is resolved into input and output irreducible ranks. In the tested system, all eleven nonzero blocks of the four quadratic maps are collinear: the residual reflects different channel weights, rather than another multiplication direction within a block. Independently, covariance composition gives a positive output-rank law for products of isotropic rank channels. Three exact Casimir moments imply double-scaled concentration and a tail bound that can control filters of at most linear deficit growth. The full memory filter and its spectral limit remain to be evaluated.

1 Rank resolution without discarding multiplicity

Use the standard-CAR, overlapping complex Gaussian p-body ensemble, with $\mathbb{E}J_{IJ}J_{KL} = w\delta_{IL}\delta_{JK}$ and $\hbar = 1$. At charge m, the scalar product is $\mathbb{E}\,\mathrm{tr}_m\, X^\dagger Y$, where tr_m is normalized. The target remains

$$\mathcal{E} = \sum_i (\mathrm{ad}_H^2\, c_i)^\dagger\, \mathrm{ad}_H^2\, c_i + \tfrac{1}{4}[H,[H,A]], \quad A = \sum_i [H,c_i]^\dagger [H,c_i]. \tag{1}$$

The preceding exact contraction gives $Q_{\mathcal{E}} = \Delta\mathcal{E}/2$ and $W_2\mathcal{E} = Q_{\mathcal{E}} - 2\mathbb{E}\mathcal{E}$. Here W_2 is coupling chaos.

Let $E_{ij} = c_i^\dagger c_j$. On neutral operators the Casimir is

$$\mathcal{C}X = \sum_{ij} [E_{ij},[E_{ji},X]], \qquad c_r = 2r(N+1-r), \quad 0 \le r \le R := \min(m, N-m). \tag{2}$$

The exact orthogonal projector is

$$P_r = \prod_{s\neq r} \frac{\mathcal{C}-c_s}{c_r-c_s}, \qquad d_r = \binom{N}{r}^2 - \binom{N}{r-1}^2. \tag{3}$$

Use the analogous projectors in the p-particle coupling space on both indices of the symmetric quadratic coefficient tensor. Together with P_r on the output they give orthogonal blocks $(a,b;r)$, with $a \le b$. The sum of their positive Gram matrices is the original Gram. Tensor-product multiplicities are retained inside each block. This is essential in the established reduced product formula of Kota–Vyas [1], Eq. (35): a triple of rank labels alone does not generally specify an intertwiner.

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2 What the finite residual actually contains

For $N = 5, p = 2, m = 3, w = 1$, resolve the four maps

$$B = H^2 - CI, \quad O = A - a_0 I, \quad J_H F = \{H, F\}/2, \quad W_2 \mathcal{E}. \tag{4}$$

Here $C = 18$, $a_0 = \mathbb{E}\,\mathrm{tr}_m\, A$, and $F = W_1 T_3 + 86H$ is the previous linear rank-filtered field. All 18 blocks are evaluated without choosing Clebsch–Gordan bases. Seven vanish. In every nonzero block the Gram of these four maps has rank one.

Define H_a by projecting the same coupling realization onto rank a. Put $Z_{aa;r} = P_r(H_a^2)$ and $Z_{ab;r} = P_r(\{H_a, H_b\})$ for $a < b$. Then the raw maps have the exact realization-wise reconstruction

$$(H^2, A, J_H F, Q_{\mathcal{E}}) = \sum_{a \le b, r} (1, o_{ab;r}, f_{ab;r}, g_{ab;r}) Z_{ab;r}. \tag{5}$$

The complete table is

(a,b)	r	$\|W_2 Z_{ab;r}\|^2$	$o_{ab;r}$	$f_{ab;r}$	$g_{ab;r}$
$(0,0)$	0	81/50	4/3	−10	128
$(0,1)$	1	864/25	4/3	−15/4	334/3
$(0,2)$	2	27	4/3	−6	352/3
$(1,1)$	0	192/25	7/4	5/2	148
$(1,1)$	1	448/75	1/2	5/2	88
$(1,1)$	2	320/3	5/4	5/2	409/4
$(1,2)$	1	160/3	2	1/4	595/4
$(1,2)$	2	224/3	1/2	1/4	647/8
$(2,2)$	0	3/2	4	−2	400
$(2,2)$	1	28/3	−1	−2	−20
$(2,2)$	2	140/3	2	−2	154

Subtracting the old three global directions leaves a positive norm because g is not one global linear combination of $1, o, f$ across all $r > 0$ blocks. Its squared norm remains 248945355/14903, agreeing exactly with the preceding calculation. Output rank zero is precisely the sample-dependent scalar part.

The rank-one assertion follows from the full coefficient Gram, not a fit to a few matrices. Two separate integer realizations also verify all eight operator identities in (5). This result does not make the representation tensor products multiplicity-free, and does not establish collinearity at general N, p, m. It identifies the source of the known finite witness.

3 A positive law for the output rank of a product

Now consider independent isotropic Hermitian rank-a and rank-b operators U, V on $\mathcal{H}_m$, normalized by $\mathbb{E}U^2 = \mathbb{E}V^2 = I$. Write $d = \binom{N}{m}$ and define

$$\pi_{ab}(r) = \mathbb{E}\|P_r(UV)\|^2_{\mathrm{tr}_m}. \tag{6}$$

These are nonnegative, sum to one, and vanish for $r > a + b$. This auxiliary product is used to control ranks; it is not being substituted for the same-coupling memory operator.

Let $\Phi_a(X) = \mathbb{E}UXU$. Its rank-$s$ eigenvalue is the known Racah crossing factor, in the conventions of our earlier covariance calculation and the terminating definition [3],

$$\chi_a(s) = {}_4F_3\left(\begin{matrix}-a, a-N-1, -s, s-N-1 \\ -m, m-N, -N-1\end{matrix}; 1\right). \tag{7}$$

Since $\mathbb{E}(UV)^\dagger X(UV) = \Phi_b\Phi_a(X)$, isotropy gives

$$\chi_a(s)\chi_b(s) = \sum_r \pi_{ab}(r)\chi_r(s). \tag{8}$$

Orthogonality of the covariance maps is equivalently

$$\sum_{s=0}^{R} d_s\chi_a(s)\chi_b(s) = \frac{d^2}{d_a}\delta_{ab}. \tag{9}$$

It follows either from the standard Racah transform or by reshuffling the orthogonal rank projectors in their Choi matrices. Therefore

$$\boxed{\pi_{ab}(r) = \frac{d_r}{d^2}\sum_{s=0}^{R} d_s\chi_a(s)\chi_b(s)\chi_r(s).} \tag{10}$$

The positivity in (10) follows from the matrix-product construction even though the displayed summands need not be positive. This is the standard covariance/Racah machinery applied to the output-rank distribution; it does not require individual multiplicity amplitudes or their phases.

For $N = 5, m = 3$, independent full CAR calculations give

$$\pi_{12} = (0, 5/27, 22/27), \qquad \pi_{22} = (1/75, 176/675, 98/135). \tag{11}$$

They agree with (10) using all 10,000 ordered coordinate pairs in each case and every output rank.

4 Exact moments and the double-scaled deficit

Set $x_r = r(N+1-r)$ and $C_f = m(N-m)(N+1)/N$. The $s=1$ case of (8) gives the exact identity

$$\mathbb{E}x_r = x_a + x_b - \frac{x_a x_b}{C_f}. \tag{12}$$

For higher moments define the monic polynomials and coefficients

$$L_j(x) = \prod_{\ell=0}^{j-1}[x - \ell(N+1-\ell)], \tag{13}$$

$$A_{nj} = \frac{(-n)_j (n-N-1)_j}{(-m)_j (m-N)_j (-N-1)_j \, j!}, \qquad \chi_r(n) = \sum_{j=0}^{n} A_{nj} L_j(x_r). \tag{14}$$

Writing $M_j = \mathbb{E}L_j(x_r)$ gives a triangular exact recursion

$$M_0 = 1, \qquad M_n = \frac{\chi_a(n)\chi_b(n) - \sum_{j<n} A_{nj} M_j}{A_{nn}}, \quad n \le R. \tag{15}$$

No unknown recoupling amplitude occurs in these summed moments.

Take $N = p^2/\lambda$, $m = \zeta N$, $a = \alpha p$, $b = \beta p$, along admissible integer sequences. Here $\lambda > 0$ is fixed, $0 \le \alpha, \beta \le 1$, and ζ stays in a compact subset of $(0,1)$. Put

$$t = a + b, \quad S = t - r, \quad D_x = x_t - x_r, \quad \theta = \alpha\beta\lambda \left[\frac{1}{\zeta(1-\zeta)} - 2\right]. \tag{16}$$

Evaluation of (15) through order three yields

$$\left(\frac{\mathbb{E}D_x}{N}, \frac{\mathbb{E}D_x^2}{N^2}, \frac{\mathbb{E}D_x^3}{N^3}\right) \longrightarrow (\theta, \theta^2 + \theta, \theta^3 + 3\theta^2 + \theta). \tag{17}$$

This follows by exact rational algebra, retaining symbolic $\alpha, \beta, \zeta, \lambda$. The denominators depend only on N, m; after scaling they are bounded away from zero on the stated compact sets. The rational degree comparison therefore also gives uniform convergence in the two input-rank fractions and filling.

Moreover $0 \le S \le t \le 2p$ and

$$D_x = S(N + 1 - 2t + S). \tag{18}$$

For $S > 0$, $D_x/(NS) = 1 + O(p/N)$ uniformly. Thus (17) also gives the first three moments of S. In particular, with fixed input deficits $a = p - u, b = p - v$ and half filling,

$$\mathbb{E}S \to 2\lambda, \qquad \operatorname{Var} S \to 2\lambda, \qquad \mathbb{E}S^3 \to 8\lambda^3 + 12\lambda^2 + 2\lambda. \tag{19}$$

These three moments agree with a Poisson law, but do not prove that law.

5 Tail control for the same-coupling quadratic problem

The preceding moments imply the useful uniform estimates

$$\limsup \mathbb{P}(S \geq K) \leq \frac{\theta^2 + \theta}{K^2}, \tag{20}$$

$$\limsup \mathbb{E}[(1+S)^2 \mathbf{1}_{S \geq K}] \leq \frac{4(\theta^3 + 3\theta^2 + \theta)}{K}, \qquad K \geq 1. \tag{21}$$

The second line uses $(1+S)^2 \leq 4S^2 \leq 4S^3/K$ on the indicated tail. It controls squared filters with at most linear growth in S.

To connect this independent-product bound to the actual Gaussian quadratic tensor, decompose $H = \sum_a H_a$ into its independent irreducible Gaussian components, with $\mathbb{E}H_a^2 = C_a I$. For $a < b$, the triangle inequality in each output-rank subspace gives

$$\mathbb{E}\|P_r\{H_a, H_b\}\|^2 \leq 4C_a C_b \pi_{ab}(r). \tag{22}$$

For $a = b$, Gaussian quadratic decoupling introduces an independent copy H_a' and gives

$$\mathbb{E}\|P_r W_2(H_a^2)\|^2 = \tfrac{1}{2}\mathbb{E}\|P_r\{H_a, H_a'\}\|^2 \tag{23}$$

$$\leq 2C_a^2 \pi_{aa}(r). \tag{24}$$

These inequalities retain the exchange interference rather than equating an ordered product with a symmetrized same-coupling field. The known input weights $C_a/C = \omega_a$ have factorial tails in the rank deficit $p - a$. Combining that input bound with (21) shows tightness of the output deficit $2p - r$ for the unfiltered $W_2(H^2)$ coefficient measure: first truncate the input deficits, then the conditional output deficit.

For the memory problem this provides a concrete conditional route. If its reduced filter amplitudes on bounded input-deficit windows are uniformly at most linear in S, then (21) controls their squared output tails by $O(K^{-1})$. The exact finite table does not yet prove this growth estimate at general N, p, m. Nor does the independent-product law determine exchange phases or multiplicity-resolved amplitudes of the full memory core.

The next calculation is therefore the general reduced filter, including multiplicity where present, followed by its weighted norm limit. Only after that step can the fourth-chaos descendants be restored and combined into the genuine Z_2 spectrum. The two finite-memory-parameter covariance, spatial reconstruction, and the bulk meson mass dictionary remain subsequent objectives.

6 Evidence, conventions and retained corrections

All numerical records are local and use the same physical conventions as the preceding contraction note. The finite tensor calculation uses $w = 1$; the product probabilities are variance-normalized. Neither is a direct assignment of de Sitter masses or finite-G dynamics.

The exact rank computation checks all projectors for idempotence, orthogonality, eigenvalue and dimension. Its 18 block Grams sum exactly to both the preceding full Gram and, after removing $r = 0$, the preceding sample-traceless Gram. The largest saved integer intermediate is 710400. Gram summation uses bounded integer chunks and rational denominators. The full coefficient tensors, not disorder samples, establish the eleven collinearity identities.

The output law is checked by four finite Racah transforms and their orthogonality relations. The factorial recursion is checked only at admissible probe orders. Two independent projected-CAR product calculations verify all output probabilities and the symmetrization bound. Eight large-p cases evaluate the exact moment formulas at 80 and 120 decimal digits; saved 65-digit upper bounds agree to the specified 10^{-60} relative threshold. These precision checks are not interval rounding certificates.

For example, at $\lambda = 1/4$, $a = b = p$, the exact-arithmetic upper bound $\mathbb{E}S^2 \le \mathbb{E}D_x^2/(N+1-4p)^2$ is approximately 0.73772646 at $p = 16$ and 0.74809847 at $p = 128$; its limit is $3/4$. The actual finite deviations are retained.

Three verification corrections are preserved with their original sources. A list was first used as a NumPy advanced index. A symbolic reconstruction initially used structural equality before expanding its exact zero residuals. A moment check initially requested probe order three when $R = 2$; its admissible domain was corrected, while the larger systems retain the full third-order check. No saved scientific result or tolerance was changed to obtain agreement.

References

[1] V. K. B. Kota and M. Vyas, *Random Matrix Theory for Transition Strength Densities in Finite Quantum Systems: Results from Embedded Unitary Ensembles*, arXiv:1504.00759, Sec. 3, Eqs. (34)–(35). The reduced product retains a multiplicity label.

[2] V. K. B. Kota, *Bivariate moments of the two-point correlation function for embedded Gaussian unitary ensemble with k-body interactions*, arXiv:2208.11312v2, Sec. IV, covariance decomposition.

[3] NIST Digital Library of Mathematical Functions, Eq. 18.26.3, terminating Racah-polynomial definition. The parameter identification is the one used in the project's earlier rank-covariance calculation.